ISAT 410 – Sustainable Energy Development

Department of Integrated Science and Technology
James Madison University

Course Description, Information, and Syllabus[1]
Spring 2024

COURSE LOCATION and TIME
EnGeo 1207, Mondays/Wednesdays 9:35–10:50 AM

INSTRUCTORS	**E-mail**	**:E LOCATION/PHONE/HOURS**
Dr. Mesude Bayrakci-Boz	bayrakmx@jmu.edu	King Hall 153, (540) 568-2530 Mondays, 11:30 AM – 1:30 PM Wednesdays 11:30 AM – 1:00 PM OR by appointment
Dr. Jonathan Miles	milesjj@jmu.edu	King Hall 155, (540) 568-3044 TBD OR by appointment

COURSE OUTLINE

Welcome to ISAT 410 – *Sustainable Energy Development*. This course addresses the nature of sustainable energy development and the science and technologies associated with solar, wind and other sustainable energy resources, and related applications e.g. solar-thermal power, photovoltaics, wind power, biomass-derived energy, and other options. Energy storage systems, combined/hybrid renewable-conventional systems for peaking and load management, passive and energy-efficient building design, and rural energy development will also be considered.

The goal of this course is to advance your understanding of energy and related technologies, especially regarding applications of renewable energy systems and sustainable energy development in the United States and beyond. The course is designed to help you to achieve the following objectives:

- Identify and explain renewable sources of energy.
- Discuss the scientific principles governing the energy content of renewable energy sources.
- Describe the component technologies involved in extracting energy from different renewable energy sources.
- Design and analyze hybrid and renewable energy systems.
- Explain the basic design and operation of energy storage systems and how they complement renewable energy sources.
- Construct optimized, conventional renewable hybrid energy systems and determine their effectiveness in load management.
- Describe the production, transportation (or transmission) and end uses of renewable energy.
- Discuss the environmental and economic issues associated with the different energy technologies and the challenges of sustainable energy development.

[1] *Information further to this syllabus pertaining to common JMU academic policies may be found at http://www.jmu.edu/syllabus/.*

It is our objective for this course to build upon concepts and skills you have learned in your sector courses and beyond. Analytical tools and concepts related to energy studies will be further developed, and methods will be applied in ISAT 410 to provide you with the ability to understand, analyze, and design renewable energy systems.

TEXTS AND MISCELLANEOUS RESOURCES

The following texts are supplemental to the course, materials from which will be incorporated into the course and may be required as reading:

- Tester, J.W. et al., 2012, SUSTAINABLE ENERGY: CHOOSING AMONG OPTIONS, Second Edition, MIT Press, Cambridge, Massachusetts.
- Dunlap, R.A., 2019, SUSTAINABLE ENERGY, Second Edition, Cengage, Boston, Massachusetts.
- Kreith, F. and R. E. West (Ed.), 1997, CRC HANDBOOK OF ENERGY EFFICIENCY, CRC Press, Inc., Florida.
- Duffie, J.A. and W. A. Beckman, 1991, SOLAR ENGINEERING OF THERMAL PROCESSES, 2nd Edition, John Wiley & Sons, Inc., New York.
- Manwell, J.F. et al., 2009, Wind Power Explained: Theory, Design and Application, U.K.
- Annual Energy Outlook 2023, U.S. Energy Information Administration.
- Solar Energy Engineering: Processes and Systems Kalogirou, 2009, Academic Press ISBN: 978-0123745019.
- Energy Resources & Systems Volume 2: Renewable Resources, by Tushar K. Ghosh, Mark A. Prelas, and Springer, 2011.
- Miscellaneous articles as announced as *required* or *optional*.

- Software provided in computer lab or or can be downloaded (not to be purchased)
    - ✓ The System Advisor Model software *developed and supported by the National Renewable Energy Laboratory (NREL) may be introduced and applied on homework and exam problems throughout the course. This is free and available to the public, you may download and become acquainted with the software at any time.* https://sam.nrel.gov/
    - ✓ Students will use HelioScope in some assignments for PV systems design. https://www.helioscope.com/

All will be made available for download at the course Canvas site.

COURSE POLICIES AND STUDENT EXPECTATIONS

1. You are expected to complete each assigned reading if any prior to the lecture to which it corresponds (see syllabus). Attendance will be taken at lectures and considered when final grades are assigned. You are responsible for all material covered and any materials handed out during lecture regardless of your attendance. To the best of your ability, please e-mail us (bayrakmx@jmu.edu and milesjj@jmu.edu) before a class if you are sick or injured and will not be attending the class. ***Understand that learning is not a passive activity – your instructor is your guide and your facilitator. We will provide you with the opportunity to gain a great deal of knowledge and understanding and enjoy the process of learning. In order to optimize your experience, a strong and consistent effort on your part is required.***

2. Each assignment must be completed by the assigned due date. You may receive credit for late work only in the case of extraordinary circumstances that must be verifiable and acceptable to your instructor. If such circumstances arise, please do not hesitate to discuss them with us. Any late submissions must be made to your instructor at the earliest opportunity possible.

3. Homework will be assigned and may be lengthy and involve new concepts. ***It is strongly advised that homework assignments not be left until the last minute, a conscientious effort on homework assignments is the key to success in this course!***
4. Students are encouraged to work together on homework assignments. However, each student must type up and submit his/her own effort. Copying of others' homework assignments is unacceptable.

Each homework assignment ***must***:

- be labeled at the beginning with **Name**, **Date**, **Collaborator**, **Assignment Number**, **Course/Section**, and **Instructor's Name**;
- be neat and legible;
- use only one side of each page of paper, and must use the appropriate type of paper;
- contain graphs and figures (when applicable) that are neat and adhere to standard practice.

Sci*entists and technology professionals are expected to present their work neatly and in a well-organized fashion.* ***Sloppy, ill-prepared, unprofessional submissions will be returned ungraded or for re-submission and with penalty applied***. ***Assignments submitted late will be penalized at the rate of one letter grade per day late***.

GRADING POLICY

Due dates for many assignments are listed on the class calendar. Updates will be provided in class and through Canvas as soon as changes occur. Weighting of the different assignments may change to better reflect the outcomes of the course. Students will be notified as soon as possible.

The various graded components of the course will contribute to the final grade as follows:

Class participation & professionalism 10%
Exams (%15 of each) 30%
Final exam 25%
Homework 35%

Any questions or concerns regarding grading of homework or quizzes must be directed to your instructor. Any questions regarding grading of exams must be addressed ***in writing*** to your instructor ***within three school days*** after the exam is returned.

The deadline for adding or dropping a Spring 2024 regular semester class through MyMadison without academic unit permission is 26th January. The deadline for adding a class with instructor and academic unit permission is5th February. The deadlines to drop a regular semester class with a "W" grade is 22nd March. No exceptions will be made to these deadlines.

MAKEUP POLICY

In general, no make-ups will be given and a grade of zero (0) will be assigned for missing homework, quizzes, exams, etc. In the event of unanticipated and extraordinary circumstances, however, exceptions can be made on a case-by-case basis. It is important to communicate your situation to your instructor as early as possible and prior to the date on which an assignment is due or an exam is to be taken.

DISABILITIES

JMU abides by Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act, which mandate reasonable accommodations be provided for students with documented disabilities. If you have not already done so, you will need to register with the Office of Disability Services, the designated office on campus to provide services for students with disabilities. The office is located in Wilson Hall, Room 107 and you may call 540-568-6705

for more information. If you have a disability and may require some type of instructional and/or examination accommodations, please contact your instructor early in the semester so that s/he can provide or facilitate provision of accommodations you may need.

UNAUTHORIZED COLLABORATION

All students are expected to comply fully with the JMU Honor Code. It is a violation of the Honor Code to collaborate in an unauthorized manner with one or more other students on an exam or on any work submitted for academic credit.

A part of the ISAT philosophy and approach is to encourage students to work with, and learn from, one another. Collaboration on homework is thus authorized, provided that it is done in the spirit of mutual learning and sharing of ideas. Any such collaboration must be documented by indicating the names of all persons with whom you collaborated on your homework. The copying of another person's work or ideas and presenting them as your own is unethical and prohibited.

All cases of copying will be reported as a potential violation of the JMU Honor Code. The penalties that can be imposed for a violation can be quite harsh. Please take the time to review the JMU Honor Code.

INCLEMENT WEATHER

If the university is closed due to bad weather, makeup work, including exams, may be scheduled at another time agreed upon by class members. Check the JMU homepage, www.jmu.edu, for information about university closures.

IN SUMMARY . . .

We encourage you to offer comments and suggestions throughout the semester. Your feedback is important and helps us to provide you the very best course possible. Welcome, once again, to ISAT 410. We look forward to an exciting and inspiring semester together!

TENTATIVE SCHEDULE

Week 1	15-Jan	Module 1	No classes
	17-Jan		**Introduction to Sustainable/Renewable Energy I**
Week 2	22-Jan		Introduction to Sustainable/Renewable Energy II
	24-Jan		Our Renewable Future
Week 3	29-Jan		Energy 101
	31-Jan		Energy Today
Week 4	5-Feb	Module 2	**Sustainable Energy Technologies**
	7-Feb		Introduction to Solar and Wind energy
Week 5	12-Feb		Hydropower Systems
	14-Feb		Hydropower Systems
Week 6	19-Feb		Ocean Energy
	21-Feb		Biomass Energy
Week 7	26-Feb		Geothermal Energy
	28-Feb		Nuclear Energy
Week 8	4-Mar		Open
	6-Mar		Exam 1
Week 9	11-Mar	SPRING BREAK	
	13-Mar		
Week 10	18-Mar	Module 3	**Solar Energy**
	20-Mar		Sunlight Calculations
Week 11	25-Mar		Sun chart and shading
	27-Mar		Designing solar system
Week 12	1-Apr		Designing solar system
	3-Apr		Analyzing Solar System
Week 13	8-Apr		Open
	10-Apr	Module 4	**Wind Energy (Overview)**
Week 14	15-Apr		Wind Measurement and Resource Analysis
	17-Apr		Wind Project Siting and Performance
Week 15	22-Apr		Exam 2
	24-Apr		Wind Energy Physics and Technologies
Week 16	29-Apr		Wind Project Development – Land
	1-May		Wind Project Development – Offshore
	FINAL EXAM WEEK		